**DAYANANDA SAGAR COLLEGE OF ENGINEERING**
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560078
(An Autonomous Institution affiliated to VTU, Belagavi)

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**Introduction to AI and Applications**

**CIA-2 Question Bank (Common to CSE/CV Streams)**

1. What is **Natural Language Processing (NLP)**? Explain how virtual assistants such as Siri and Alexa apply NLP techniques to understand and respond to user queries.
2. List five application areas where **AI-driven robotic systems** are used. Explain the benefits and improvements they offer in each of these fields.
3. Explain how **spam filtering** works as an application of NLP. What methods are used by systems to classify an email as spam or legitimate?
4. Describe how machine vision differs from human eyesight. Give two real-world applications of machine vision.
5. State **Mitchell's definition of Machine Learning**. Describe the roles of **Task (T)**, **Experience (E)**, and **Performance (P)**, providing one suitable example for each.
6. Provide an overview of **low-code and no-code AI platforms**, highlighting their major advantages and limitations. Support your answer with relevant examples.
7. Analyse why the gap between AI experts and domain experts makes no-code AI essential for modern organisations. Support your answer with examples.
8. Discuss two practical applications of **supervised learning**—one that involves **classification** and another that involves **regression**—with real-world examples.
9. Examine how **computer vision, image recognition, and deep learning** work together in self-driving cars. Analyze the potential safety risks if any one of these components fails.
10. Differentiate between the steps involved in traditional **machine learning model development** and those used in **no-code AI platforms**. How do no-code tools simplify and accelerate AI development for organizations?
11. Explain why **semi-supervised learning** is considered an intermediate approach between supervised and unsupervised learning. In which real-world situations is it more advantageous than fully supervised methods?
12. Explain the two main techniques used in **semi-supervised learning** to develop models using both labeled and unlabeled datasets.
13. Describe the four major types of **unsupervised learning techniques**, providing one suitable example for each category.

14. Compare **reinforcement learning (RL)** with supervised learning. Discuss why RL is commonly used in areas like gaming and robotics, and outline its main challenges. Why does RL depend heavily on large volumes of simulated data?
15. Using a **robot-and-diamond scenario**, explain how **reinforcement learning (RL)** works. Describe how the robot learns to reach the diamond while avoiding obstacles like fire. Clearly define the **state, actions, and reward/penalty system** involved.
16. Evaluate the importance of **RMSE** and **Adjusted $R^2$** as performance metrics in machine learning models. How does **k-fold cross-validation** address the drawbacks of a simple train-test split? Describe the five steps involved in performing k-fold cross-validation.
17. Explain how to select an appropriate **clustering method** for a given problem. Discuss the commonly used clustering approaches and provide examples for each.

Along with these questions one example problem on **linear regression or KNN Classification models.**